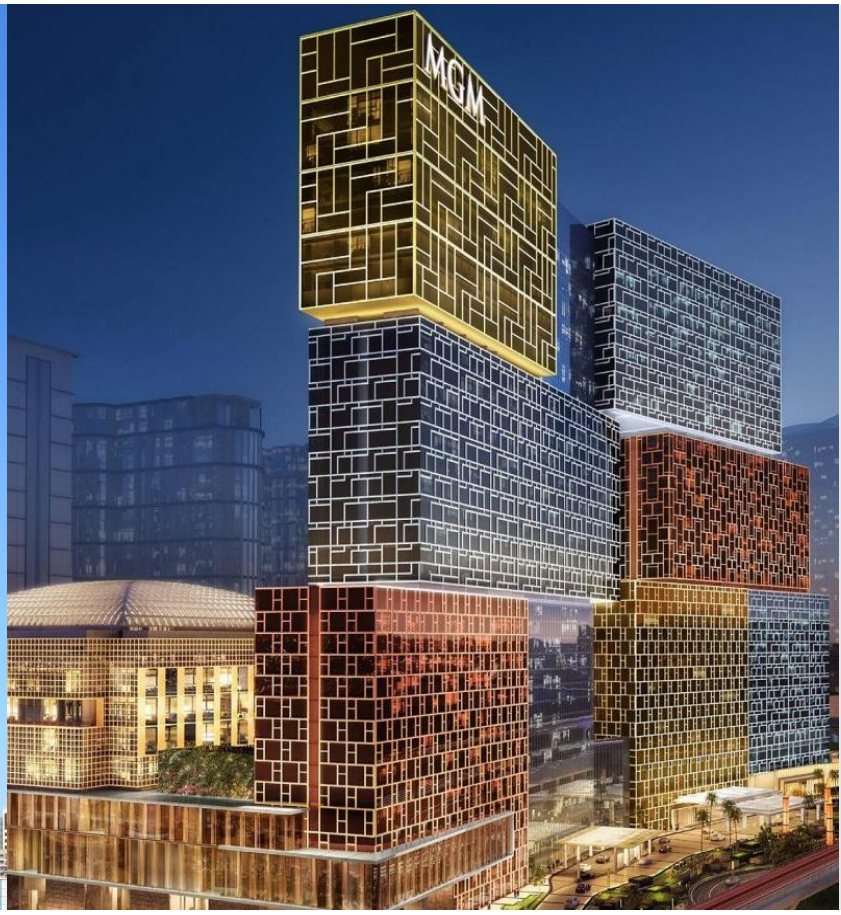
# MGM MACAU & COTAI
# RESOURCE EFFICIENCY MANAGEMENT REPORT
**DECEMBER 2023**

Prepared For:

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# EXECUTIVE SUMMARY

NV5 has been working closely with MGM to implement the EMOs as well as identify additional opportunities to achieve ISO certification energy targets. NV5 appreciates the great support and trust from MGM team over the years, in particular the Facility Management and the Sustainability Teams.

## Key Performance Indicators

NV5 reports both normalized and actual savings on a monthly basis for both properties. The normalized savings are determined based on a 2019 baseline, with aligned annual savings targets to meet both ISO50001 as well as NV5's retainer contract requirements. Please note that the reported normalized savings achieved are to-date and cumulative for each month of the respective year.

*Table 1. ISO and Retainer Contract Energy Performance Targets for 2025*

| MGM Properties | Baseline | Energy Savings | Normalized Savings |
|---|---|---|---|
| MACAU | 2019 | 2025 Target | **-13%** |
| | | Savings as of Dec 2023 | **-10.4%** |
| COTAI | 2019 | 2025 Target | **-13%** |
| | | Savings as of Dec 2023 | **-5.7%** |

The actual monthly savings are determined against the 2019 baseline consumption and cumulative for each month of 2023.

*Table 2. Internal Actual Energy Performance Targets for 2025*

| MGM Properties | Baseline | Actual Saving Internal Target[1] | Savings Achieved Up to December 2023 |
|---|---|---|---|
| MACAU | 2019 | 13% | **-1.5%** |
| COTAI | 2019 | 13% | **-1.5%** |

[1] Actual Energy Savings Target for 2025 has been established based on future CAPEX projects, following review with MGM teams.

## Highlighted Changes in Operation

### MACAU

1. New water-to-water HX was installed for podium MVAC space heating, testing works were conducted since Feb 2023.
2. Grande Praça 7 Units AHU were installed with PICCV.
3. MGM MACAU implemented the Hotel Tower Staircase lighting delamping by removing one (1) piece of T5 tube from the existing fluorescent lighting fixture.
4. MGM MACAU replaced 6 units of UPS with high efficiency model.
5. MGM MACAU will replace the rooftop façade LED by Q1 2024.
6. MGM MACAU will upgrade the Lion Backdrop lighting by Q1 2024.
7. MGM MACAU have applied the N+2 Cooling Tower with lower fan speed control logic to address the noise concerns from One Central.
8. MGM MACAU will further fine tune the non-Casino AHU fresh air damper, temperature setpoint, fan speed for energy saving.
9. Energenz is working with the project team to minimize the energy increase from IC/GIC/LPK additional equipment but maintain the new thermal comfort design requirements.
10. MGM MACAU switched ON the Main Steam Boiler from 15 – 27 of December 2023 due to OAT drop below 15°C. The Main Steam Boiler operation stopped on 28 December 2023.

### COTAI

1. Grill 58, MDM, Ballroom, Hao Guo, Coast kitchen exhaust fan steps speed control program and graphics were updated in BMS.
2. MGM COTAI hotel towers façade lighting control has transferred to IT team, it will be dimmed down from 02:00 – 17:30 instead of existing switch OFF. This is to reduce the LED failure rate and extend the LED life time.
3. MGM COTAI Mansion AHU-6FM-01, 03, 08 has utilized the LoRa sensor data to reset the RAT setpoint.
4. MGM COTAI installed additional fourteen (14) sets of charging stations in car park.
5. MGM COTAI Kitchen Exhaust Fan operation hours have been fine-tuned according to the latest F&B team operation needs.
6. MGM COTAI adjusted the settings of AH-3F-1 exhaust fans, fresh air damper position, SAT and VSD speed for the GF Lotus Gaming Area during the Lucky Draw Period.
7. MGM COTAI completed 8 units of high efficiency centralized RO / booster pump replacement.
8. MGM COTAI back of house office lighting panel upgrade with occupancy sensor was completed.
9. MGM COTAI has adopted the passive radiative cooling paint on the North Tower rooftop ELV room as trial.

### Consumption Intensity

The 2023 YTD electricity and water use intensities are tracked on a monthly basis for both properties:

*Table 3. Year to Date Actual Electricity Use Intensity (EUI)*

| MGM Properties | YTD EUI (kWh/m²) | % Change Compared to 2022 |
|---|---|---|
| MACAU | **27.0** | **+10.3%** |
| COTAI | **27.7** | **+14.7%** |

*Table 4. Year to Date Actual Water Use Intensity (WUI)*

| MGM Properties | YTD WUI (m³/ m²) | % Change Compared to 2022 |
|---|---|---|
| MACAU | **0.27** | **+39.6%** |
| COTAI | **0.20** | **+55.8%** |

## ISO 50001 Certification Audit

### MACAU

The HKQAA 2023 surveillance audit was completed on 28 April 2023. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

### COTAI

The HKQAA 2023 surveillance audit was completed on 7 December 2023. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

## Milestones in Year 2023

The following activities are completed in year 2023:

*Table 5. MGM MACAU Year 2023 Projects*

| Item | Initiatives | Status |
|---|---|---|
| **Electricity** | | |
| 1 | Secondary Chilled Water Pump Retrofit (Phase 1) | Postpone to 2024 |
| 2 | Cooling Tower Fan Retrofit (Phase 1) | On-going |
| 3 | PICCV Implementation 2023 (Grande Praça 7 Units) | Completed |
| 4 | Neon Light LED Replacement (Phase 3) | Postpone to 2024 |
| 5 | Hotel Tower Staircase Lighting Delamping | Completed |
| 6 | Rooftop Façade LED Replacement | On-going |
| 7 | Lion Backdrop Lighting Upgrade | On-going |
| **LPG** | | |
| 8 | Natural Gas Changeover Project | On-going |

*Table 6. MGM COTAI Year 2023 Projects*

| Item | Initiatives | Status |
|---|---|---|
| **Electricity** | | |
| 1 | Chiller Plant Controls Optimization (CHWP DP Setpoint Reset, CHWST Setpoint Reset) | On-going |
| 2 | PICCV Implementation 2023 (12 Units) | Postpone to 2024 |
| 3 | KEF Steps Speed Control Optimization<br>(Vista, Ball Room, MDM, Hao Guo, Five Foot Road, Chún, Coast) | Completed |
| 4 | AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2) | On-going |
| 5 | Centralized RO Pump Set Replacement (6 units) | Completed |
| 6 | Booster Pump Set Replacement (2 units) | Completed |
| **Water** | | |
| 7 | Extension of B2 Carpark CDP Water Recycling (Phase 5 to 7) | Completed |
| 8 | Extension of B2 Carpark CDP Water Recycling (Phase 11) | On-going |

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# ACKNOWLEDGMENT

NV5 has been working with MGM Team as the energy and sustainability manager to achieve energy savings in MGM properties and feed operational intelligence into the design.

The major goal of this project is not only to realize operation energy savings but also to extend the life of equipment, hence reducing CAPEX. MGM has demonstrated continuous commitment to be an environmentally and socially responsible business in the past years. In addition to that NV5 has assisted MGM MACAU and COTAI to achieve several other energy management and sustainability awards and certifications, such as ISO 50001, CEM Energy Savings Award, Macao Green Hotel Award and China 3 Star Green Building Operation Label.

NV5 appreciates the great support and trust from MGM Team over the years, especially the Facilities Management and the Sustainability Teams.

# 1.0 PORTFOLIO PERFORMANCE SUMMARY

## 1.1 ELECTRICITY USE INTENSITY (EUI)

MACAU's electricity use intensity (EUI) in December 2023 had increased by **14.6%** compared to the same month in 2022 and increased by **10.3%** in 2023 compared to 2022. Meanwhile, COTAI's EUI in December 2023 had increased by **18.4%** compared to the same month in 2022 and increased by **14.7%** in 2023 compared to 2022.

*Table 7. Actual Electricity Consumption Intensity (EUI) Comparison*

| Monthly Period | MACAU | | | COTAI | | |
|---|---|---|---|---|---|---|
| | Year 2022 Actual EUI (kWh/m²) | Year 2023 Actual EUI (kWh/m²) | % Change | Year 2022 Actual EUI (kWh/m²) | Year 2023 Actual EUI (kWh/m²) | % Change |
| **January** | 22.6 | 22.4 | -0.8% | 22.3 | 22.6 | +1.6% |
| **February** | 20.2 | 21.2 | +5.2% | 20.1 | 21.0 | +4.5% |
| **March** | 24.4 | 24.6 | +1.0% | 23.6 | 24.4 | +3.3% |
| **April** | 24.3 | 25.0 | +3.0% | 24.0 | 25.8 | +7.3% |
| **May** | 25.9 | 28.5 | +9.9% | 25.6 | 29.4 | +14.7% |
| **June** | 27.3 | 30.3 | +11.1% | 27.0 | 31.5 | +17.0% |
| **July** | 25.9 | 31.9 | +23.1% | 24.3 | 32.5 | +33.7% |
| **August** | 28.1 | 31.6 | +12.7% | 26.3 | 32.0 | +21.5% |
| **September** | 26.8 | 30.3 | +13.4% | 25.6 | 30.5 | +19.2% |
| **October** | 24.6 | 28.7 | +17.0% | 24.4 | 29.2 | +20.0% |
| **November** | 22.7 | 25.2 | +11.0% | 23.3 | 25.8 | +11.0% |
| **December** | 21.2 | 24.3 | +14.6% | 21.0 | 24.9 | +18.4% |
| **Average** | **24.5** | **27.0** | **+10.3%** | **24.0** | **27.5** | **+14.7%** |

## 1.2 WATER USE INTENSITY (WUI)

MACAU's water use intensity (WUI) in December 2023 has increased by **47.6%** compared to the same month in 2022 and increased by **39.6%** in 2023 compared to 2022. Meanwhile, COTAI's WUI in December 2023 has increased by **56.8%** compared to the same month in 2022 and increased by **55.8%** in 2023 compared to 2022.

*Table 8. Actual Water Consumption Intensity (WUI) Comparison*

| Monthly Period | MACAU | | | COTAI | | |
|---|---|---|---|---|---|---|
| | Year 2022 Actual WUI ($m^3/m^2$) | Year 2023 Actual WUI ($m^3/m^2$) | % Change | Year 2022 Actual WUI ($m^3/m^2$) | Year 2023 Actual WUI ($m^3/m^2$) | % Change |
| **January** | 0.18 | 0.19 | +6.0% | 0.12 | 0.14 | +21.0% |
| **February** | 0.20 | 0.21 | +8.6% | 0.12 | 0.16 | +35.4% |
| **March** | 0.19 | 0.21 | +11.9% | 0.12 | 0.17 | +37.9% |
| **April** | 0.21 | 0.24 | +10.3% | 0.14 | 0.19 | +36.7% |
| **May** | 0.23 | 0.28 | +20.1% | 0.15 | 0.21 | +44.2% |
| **June** | 0.19 | 0.29 | +51.9% | 0.12 | 0.23 | +84.2% |
| **July** | 0.15 | 0.32 | +118.6% | 0.10 | 0.25 | +149.5% |
| **August** | 0.19 | 0.34 | +73.3% | 0.13 | 0.23 | +74.1% |
| **September** | 0.22 | 0.30 | +37.3% | 0.15 | 0.22 | +49.0% |
| **October** | 0.19 | 0.30 | +58.8% | 0.14 | 0.21 | +54.5% |
| **November** | 0.19 | 0.30 | +54.9% | 0.14 | 0.21 | +47.6% |
| **December** | 0.17 | 0.25 | +47.6% | 0.11 | 0.17 | +56.8% |
| **Average** | **0.19** | **0.27** | **+39.6%** | **0.13** | **0.20** | **+55.8%** |

# 2.0 MGM MACAU PERFORMANCE

## 2.1 ELECTRICITY CONSUMPTION

The actual electricity consumption and cost in December 2023 increased by 14.6% and 12.8% respectively compared to the same month in 2022. To date, MGM MACAU has increased the actual electricity consumption by 10.3% and associated cost increased by 9.8%.

*Table 9. MACAU Summary of Actual Monthly Electricity Consumption and Cost*

| Monthly Period | Actual Electricity Consumption (kWh) | | | Actual Electricity Cost (MOP) | | | % Change – CDD |
|---|---|---|---|---|---|---|---|
| | 2022 | 2023 | % Change Actual | 2022 | 2023 | % Change Actual | |
| **January** | 4,784,280 | 4,745,400 | -0.8% | 5,519,350 | 5,668,020 | +2.7% | -37.7% |
| **February** | 4,266,240 | 4,488,360 | +5.2% | 4,942,590 | 5,365,140 | +8.5% | +658.0% |
| **March** | 5,152,320 | 5,205,240 | +1.0% | 5,938,160 | 6,207,870 | +4.5% | -9.0% |
| **April** | 5,131,680 | 5,284,800 | +3.0% | 6,183,420 | 6,236,780 | +0.9% | -3.2% |
| **May** | 5,478,840 | 6,019,800 | +9.9% | 6,595,030 | 7,106,470 | +7.8% | +17.6% |
| **June** | 5,777,040 | 6,418,080 | +11.1% | 7,567,480 | 8,307,460 | +9.8% | +6.6% |
| **July** | 5,480,880 | 6,748,440 | +23.1% | 7,172,780 | 8,730,530 | +21.7% | -3.1% |
| **August** | 5,938,920 | 6,692,280 | +12.7% | 7,799,020 | 8,647,480 | +10.9% | +3.7% |
| **September** | 5,659,200 | 6,417,720 | +13.4% | 7,450,220 | 8,297,960 | +11.4% | -9.6% |
| **October** | 5,194,440 | 6,075,360 | +17.0% | 6,209,630 | 7,129,400 | +14.8% | +3.7% |
| **November** | 4,810,800 | 5,339,640 | +11.0% | 5,737,570 | 6,263,910 | +9.2% | +3.2% |
| **December** | 4,494,480 | 5,149,080 | +14.6% | 5,360,740 | 6,045,880 | +12.8% | +2461.8% |
| **Total/Average** | **62,169,120** | **68,584,200** | **+10.3%** | **76,475,990** | **84,006,900** | **+9.8%** | **+5.7%** |

***Important note**: Cooling Degree Days (CDDs) is the main variable taken into consideration on electricity consumption normalization.*

NV5 summarized the key items that could result in an electricity consumption increase below:

- Chiller Plant electricity consumption has increased by 46.1% in December 2023 compared with the same month in 2022. The outdoor average temperature and cooling degree days have increased 21.6% and 2461.8% respectively in December 2023.

Table 10 presents the comparison of electricity rates in the years 2022 and 2023. The electricity rate in December 2023 decreased by **1.6%** compared to the same month in 2022.

*Table 10. MACAU - Summary of Actual Monthly Electricity Rate*

| Monthly Period | Year 2022 – Electricity Rate (MOP/kWh) | Year 2023 – Electricity Rate (MOP/kWh) | % Change – Electricity Rate |
|---|---|---|---|
| **January** | 1.15 | 1.19 | +3.5% |
| **February** | 1.16 | 1.20 | +3.2% |
| **March** | 1.15 | 1.19 | +3.5% |
| **April** | 1.20 | 1.18 | -2.1% |
| **May** | 1.20 | 1.18 | -1.9% |
| **June** | 1.31 | 1.29 | -1.2% |
| **July** | 1.31 | 1.29 | -1.1% |
| **August** | 1.31 | 1.29 | -1.6% |
| **September** | 1.32 | 1.29 | -1.8% |
| **October** | 1.20 | 1.17 | -1.8% |
| **November** | 1.19 | 1.17 | -1.6% |
| **December** | 1.19 | 1.17 | -1.6% |
| **Total/Average** | **1.23** | **1.22** | **-0.4%** |

Table 11 summarizes the monthly normalized electricity consumption. The actual electricity consumption in December 2023 decreased by **22.4%** compared to 2019 normalized consumption in the same month.

*Table 11. MACAU - Summary of Normalized Monthly Electricity Consumption*

| Monthly Period | 2019 Normalized Electricity Consumption (kWh) | 2023 Actual Electricity Consumption (kWh) | % Change – Normalized |
|---|---|---|---|
| **January** | 4,688,691 | 4,745,400 | +1.2% |
| **February** | 4,625,432 | 4,488,360 | -3.0% |
| **March** | 5,253,809 | 5,205,240 | -0.9% |
| **April** | 5,711,541 | 5,284,800 | -7.5% |
| **May** | 6,338,759 | 6,019,800 | -5.0% |
| **June** | 6,851,234 | 6,418,080 | -6.3% |
| **July** | 7,847,650 | 6,748,440 | -14.0% |
| **August** | 7,949,285 | 6,692,280 | -15.8% |
| **September** | 7,552,268 | 6,417,720 | -15.0% |
| **October** | 7,538,720 | 6,075,360 | -19.4% |
| **November** | 6,680,060 | 5,339,640 | -20.1% |
| **December** | 6,635,038 | 5,149,080 | -22.4% |
| **Total/Average** | **77,672,486** | **68,584,200** | **-11.7%** |

## 2.2 LPG CONSUMPTION

LPG consumption can be broken down into three (3) categories: boiler, hotel upper zone DHW, and kitchens/restaurants. Since the boiler and restaurants LPG utilization cover over 95% of the overall site LPG consumption, NV5 focuses the analysis on those two areas.

According to Table 12, the actual LPG consumption in December 2023 increased by **5.7%** while the cost increased by **24.8%** respectively compared to the same month in 2022. To date, MGM MACAU has decreased the actual LPG consumption by **6.2%** and associated costs decreased by **12.0%.**

*Table 12. MACAU – Summary of Actual Monthly LPG Consumption and Cost*

| Monthly Period | Actual LPG Consumption (kg) | | | Actual LPG Cost (MOP) | | | % Change – Min OAT |
|---|---|---|---|---|---|---|---|
| | 2022 | 2023 | % Change Actual | 2022 | 2023 | % Change Actual | |
| **January** | 121,221 | 123,108 | +1.6% | 1,327,490 | 1,084,579 | -18.3% | -13.7% |
| **February** | 146,623 | 78,483 | -46.5% | 1,450,103 | 768,347 | -47.0% | +105.6% |
| **March** | 91,040 | 77,597 | -14.8% | 900,382 | 759,672 | -15.6% | +5.7% |
| **April** | 85,743 | 77,139 | -10.0% | 858,756 | 755,191 | -12.1% | +42.3% |
| **May** | 82,704 | 80,383 | -2.8% | 992,447 | 839,195 | -15.4% | +15.4% |
| **June** | 77,044 | 71,232 | -7.5% | 924,531 | 743,662 | -19.6% | +2.8% |
| **July** | 44,991 | 67,441 | +49.9% | 539,887 | 704,081 | +30.4% | +2.7% |
| **August** | 67,705 | 71,984 | +6.3% | 697,361 | 625,540 | -10.3% | +4.7% |
| **September** | 72,270 | 69,953 | -3.2% | 744,377 | 607,893 | -18.3% | -0.8% |
| **October** | 80,109 | 78,004 | -2.6% | 825,127 | 677,857 | -17.8% | +3.6% |
| **November** | 74,805 | 77,482 | +3.6% | 659,028 | 805,816 | +22.3% | -6.0% |
| **December** | 110,651 | 117,003 | +5.7% | 974,840 | 1,216,832 | +24.8% | -16.7% |
| **Total/Average** | **1,054,905** | **989,808** | **-6.2%** | **10,894,328** | **9,588,665** | **-12.0%** | **+7.2%** |

***Important note:*** *Minimum Outdoor Air Temperature (OAT) is the main variable taken into consideration on LPG consumption normalization.*

NV5 summarized the key items which could result in LPG consumption increase below:

- Kitchen LPG consumption for December 2023 has increased by 24.1% compared with the same month in 2022. Food cover has increased by 223.4% in December 2023.
- Boiler LPG consumption for December 2023 has increased by 14.0% compared with the same month in 2022.

Table 13 presents the comparison of LPG rates in the years 2022 and 2023. The LPG rate in December 2023 increased by 18.0% compared with the same month in 2022.

*Table 13. MACAU - Summary of Actual Monthly LPG Rate*

| Monthly Period | Year 2022 – LPG Rate (MOP/kg) | Year 2023 – LPG Rate (MOP/kg) | % Change – LPG Rate |
|---|---|---|---|
| **January** | 10.95 | 8.81 | -19.6% |
| **February** | 9.89 | 9.79 | -1.0% |
| **March** | 9.89 | 9.79 | -1.0% |
| **April** | 10.02 | 9.79 | -2.3% |
| **May** | 12.00 | 10.44 | -13.0% |
| **June** | 12.00 | 10.44 | -13.0% |
| **July** | 12.00 | 10.44 | -13.0% |
| **August** | 10.30 | 8.69 | -15.6% |
| **September** | 10.30 | 8.69 | -15.6% |
| **October** | 10.30 | 8.69 | -15.6% |
| **November** | 8.81 | 10.40 | +18.0% |
| **December** | 8.81 | 10.40 | +18.0% |
| **Total/Average** | **10.44** | **9.70** | **-7.1%** |

Table 14 summarizes the monthly normalized LPG consumption. The actual LPG consumption in December 2023 has increased by 15.2% compared to 2019 normalized consumption in the same month.

*Table 14. MACAU - Summary of Normalized Monthly LPG Consumption*

| Monthly Period | 2019 Normalized LPG Consumption (kg) | 2023 Actual LPG Consumption (kg) | % Change – Normalized |
|---|---|---|---|
| **January** | 103,641 | 123,108 | +18.8% |
| **February** | 84,783 | 78,483 | -7.4% |
| **March** | 77,414 | 77,597 | +0.2% |
| **April** | 76,016 | 77,139 | +1.5% |
| **May** | 80,862 | 80,383 | -0.6% |
| **June** | 82,361 | 71,232 | -13.5% |
| **July** | 81,955 | 67,441 | -17.7% |
| **August** | 88,801 | 71,984 | -18.9% |
| **September** | 78,246 | 69,953 | -10.6% |
| **October** | 75,988 | 78,004 | +2.7% |
| **November** | 81,212 | 77,482 | -4.6% |
| **December** | 101,584 | 117,003 | +15.2% |
| **Total/Average** | **1,012,864** | **989,808** | **-2.3%** |

## 2.3 ISO50001 ENERGY CONSUMPTION RECORD

The MACAU ISO50001 energy performance target is to achieve 13% normalized energy savings by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to December 2023 is **10.4%** as shown in Table 15. Table 16 below summarizes the site monthly energy consumption (Electricity and LPG combined) compared to the 2019 baseline.

*Table 15. MACAU ISO 50001 Normalized Energy Performance Targets for 2025*

| MGM Properties | Baseline | Energy Savings | Normalized Savings |
|---|---|---|---|
| MACAU | 2019 | 2025 Target | **-13%** |
| | | Savings as of Dec 2023 | **-10.4%** |

*Table 16. MACAU ISO 50001 Normalized Monthly Energy Consumption*

| Monthly Period | 2019 Normalized Energy Consumption (GJ) | 2023 Actual Energy Consumption (GJ) | % Change – Normalized |
|---|---|---|---|
| **January** | 21,574 | 22,660 | +5.0% |
| **February** | 20,492 | 19,713 | -3.8% |
| **March** | 22,421 | 22,254 | -0.7% |
| **April** | 24,005 | 22,520 | -6.2% |
| **May** | 26,483 | 25,313 | -4.4% |
| **June** | 28,395 | 26,332 | -7.3% |
| **July** | 31,964 | 27,349 | -14.4% |
| **August** | 32,640 | 27,353 | -16.2% |
| **September** | 30,733 | 26,273 | -14.5% |
| **October** | 30,582 | 25,405 | -16.9% |
| **November** | 27,727 | 22,733 | -18.0% |
| **December** | 28,488 | 23,837 | -16.3% |
| **Total/Average** | **325,504** | **291,741** | **-10.4%** |

***Important note:*** *Factors used for unit conversion: Electricity (0.0036) and LPG (0.0453)*

## 2.4 WATER CONSUMPTION

Based on the existing water metering system, the overall site water consumption can be broken down into five (5) categories: Kitchen and Restaurants; Hotel Tower; Fire Services; Chiller Plant & miscellaneous.

According to Table 17, the actual water consumption and cost in December 2023 has increased by **47.6%** and **47.5%** respectively compared to the same month in 2022 respectively. To date, MGM MACAU has increased both the actual water consumption and associated cost by **39.6%** and **39.5%** respectively.

*Table 17. MACAU - Summary of Actual Monthly Water Consumption and Cost*

| Monthly Period | Actual Water Consumption ($m^3$) | | | Actual Water Cost (MOP) | | | Normalization Factors | |
|---|---|---|---|---|---|---|---|---|
| | 2022 | 2023 | % Change Actual | 2022 | 2023 | % Change Actual Cost | % Change – CDD | % Change – Food Covers |
| **January** | 37,078 | 39,305 | +6.0% | 287,500 | 305,020 | +6.1% | -37.7% | +52.7% |
| **February** | 41,524 | 45,112 | +8.6% | 322,270 | 350,080 | +8.6% | +658.0% | +16.6% |
| **March** | 39,962 | 44,725 | +11.9% | 310,160 | 347,050 | +11.9% | -9.0% | +71.8% |
| **April** | 45,295 | 49,960 | +10.3% | 351,540 | 387,650 | +10.3% | -3.2% | +87.5% |
| **May** | 49,042 | 58,878 | +20.1% | 380,590 | 456,800 | +20.0% | +17.6% | +77.2% |
| **June** | 40,383 | 61,331 | +51.9% | 313,390 | 475,780 | +51.8% | +6.6% | +195.1% |
| **July** | 31,179 | 68,165 | +118.6% | 242,100 | 528,630 | +118.4% | -3.1% | +2472.4% |
| **August** | 41,073 | 71,191 | +73.3% | 318,730 | 552,160 | +73.2% | +3.7% | +256.6% |
| **September** | 46,713 | 64,118 | +37.3% | 362,350 | 497,260 | +37.2% | -9.6% | +179.5% |
| **October** | 40,566 | 64,415 | +58.8% | 314,790 | 499,470 | +58.7% | +3.7% | +167.5% |
| **November** | 40,744 | 63,109 | +54.9% | 316,260 | 489,490 | +54.8% | +3.2% | +159.0% |
| **December** | 35,721 | 52,733 | +47.6% | 277,280 | 409,050 | +47.5% | +2462% | +223.4% |
| **Total/Average** | **489,280** | **683,042** | **+39.6%** | **3,796,960** | **5,298,440** | **+39.5%** | **+5.7%** | **+142.7%** |

***Important note:*** *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on water consumption normalization.*

NV5 summarized the key items which could result in water consumption increase below:

- Hotel water consumption for December 2023 has increased by 12.2% compared with the same month in 2022.
- Kitchen water consumption for November 2023 has increased by 82.7% compared with the same month in 2022.
- Podium water consumption for November 2023 has increased by 48.7% compared with the same month in 2022.

Table 18 presents the comparison of water rates in the years 2022 and 2023. The water rate in December 2023 has decreased by 0.07% compared to the same month in 2022. The average utility rate of water in 2023 has decreased by 0.04% compared to 2022.

*Table 18. MACAU - Summary of Actual Monthly Water Rates*

| Monthly Period | Year 2022 – Water Rate (MOP/m³) | Year 2023 – Water Rate (MOP/m³) | % Change – Water Rate |
|---|---|---|---|
| **January** | 7.754 | 7.760 | +0.08% |
| **February** | 7.761 | 7.760 | -0.01% |
| **March** | 7.761 | 7.760 | -0.02% |
| **April** | 7.761 | 7.759 | -0.02% |
| **May** | 7.760 | 7.758 | -0.03% |
| **June** | 7.760 | 7.758 | -0.04% |
| **July** | 7.765 | 7.755 | -0.12% |
| **August** | 7.760 | 7.756 | -0.05% |
| **September** | 7.757 | 7.755 | -0.02% |
| **October** | 7.760 | 7.754 | -0.08% |
| **November** | 7.762 | 7.756 | -0.08% |
| **December** | 7.762 | 7.757 | -0.07% |
| **Total/Average** | **7.760** | **7.757** | **-0.04%** |

Table 19 summarized the monthly normalized water consumption. The actual water consumption in December 2023 has decreased by 11.8% compared to 2019 normalized consumption in the same month.

*Table 19. MACAU - Summary of Normalized Monthly Water Consumption*

| Monthly Period | 2019 Normalized Water Consumption (m³) | 2023 Actual Water Consumption (m³) | % Change – Normalized |
|---|---|---|---|
| **January** | 43,437 | 39,305 | -9.5% |
| **February** | 42,590 | 45,112 | +5.9% |
| **March** | 47,450 | 44,725 | -5.7% |
| **April** | 50,917 | 49,960 | -1.9% |
| **May** | 55,459 | 58,878 | +6.2% |
| **June** | 59,235 | 61,331 | +3.5% |
| **July** | 67,617 | 68,165 | +0.8% |
| **August** | 68,646 | 71,191 | +3.7% |
| **September** | 65,468 | 64,118 | -2.1% |
| **October** | 65,936 | 64,415 | -2.3% |
| **November** | 59,366 | 63,109 | +6.3% |
| **December** | 59,820 | 52,733 | -11.8% |
| **Total/Average** | **685,941** | **683,042** | **-0.4%** |

## 2.5 MACAU YEAR 2023 PROJECTS

NV5 had performed analysis and coordinate with MGM to implement the following projects in 2023 and reduce the site energy and water consumption which supports NV5 annual normalized energy savings target.

*Table 20. MGM MACAU Year 2023 Projects*

| Item | Initiatives | Status |
|---|---|---|
| **Electricity** | | |
| 1 | Secondary Chilled Water Pump Retrofit (Phase 1) | Postpone to 2024 |
| 2 | Cooling Tower Fan Retrofit (Phase 1) | On-going |
| 3 | PICCV Implementation 2023 (Grande Praça 7 Units) | Completed |
| 4 | Neon Light LED Replacement (Phase 3) | Postpone to 2024 |
| 5 | Hotel Tower Staircase Lighting Delamping | Completed |
| 6 | Rooftop Façade LED Replacement | On-going |
| 7 | Lion Backdrop Lighting Upgrade | On-going |
| **LPG** | | |
| 8 | Natural Gas Changeover Project | Postpone to 2024 |

# 3.0 MGM COTAI PERFORMANCE

## 3.1 ELECTRICITY CONSUMPTION

The actual electricity consumption and cost in December 2023 increased by 18.4% and 16.7% respectively compared to the same month in 2022. To date, MGM COTAI has increased the actual electricity consumption by 14.7% and associated cost increased by 14.0%.

*Table 21. COTAI Summary of Actual Monthly Electricity Consumption and Cost*

| Monthly Period | Actual Electricity Consumption (kWh) | | | Actual Electricity Cost (MOP) | | | % Change – CDD |
|---|---|---|---|---|---|---|---|
| | 2022 | 2023 | % Change Actual | 2022 | 2023 | % Change Actual | |
| **January** | 8,336,400 | 8,471,280 | +1.6% | 10,013,460 | 10,516,100 | +5.0% | -37.7% |
| **February** | 7,528,560 | 7,869,600 | +4.5% | 9,066,060 | 9,794,150 | +8.0% | +658.0% |
| **March** | 8,833,200 | 9,128,520 | +3.3% | 10,632,040 | 11,334,370 | +6.6% | -9.0% |
| **April** | 8,985,360 | 9,638,280 | +7.3% | 11,262,370 | 11,855,350 | +5.3% | -3.2% |
| **May** | 9,585,000 | 10,993,680 | +14.7% | 11,983,070 | 13,528,490 | +12.9% | +17.6% |
| **June** | 10,087,080 | 11,802,600 | +17.0% | 12,615,080 | 14,491,170 | +14.9% | +6.6% |
| **July** | 9,082,680 | 12,146,160 | +33.7% | 11,346,140 | 14,926,180 | +31.6% | -3.1% |
| **August** | 9,847,320 | 11,961,480 | +21.5% | 12,318,540 | 14,684,180 | +19.2% | +3.7% |
| **September** | 9,587,280 | 11,428,920 | +19.2% | 12,002,380 | 14,051,860 | +17.1% | -9.6% |
| **October** | 9,119,160 | 10,939,680 | +20.0% | 11,330,350 | 13,353,730 | +17.9% | +3.7% |
| **November** | 8,706,120 | 9,667,440 | +11.0% | 10,825,450 | 11,823,780 | +9.2% | +3.2% |
| **December** | 7,875,120 | 9,325,680 | +18.4% | 9,769,740 | 11,398,440 | +16.7% | +2461.8% |
| **Total/Average** | **107,573,280** | **123,373,320** | **+14.7%** | **133,164,680** | **151,757,800** | **+14.0%** | **+5.7%** |

***Important note**: Cooling Degree Days (CDDs) is the main variable taken into consideration on electricity consumption normalization.*

NV5 summarized the key items which could result in electricity consumption increase below:

- Chiller Plant electricity consumption for December 2023 has increased by 111.4% compared to the same month in 2022. The outdoor average temperature and cooling degree days have increased 21.6% and 2461.8% respectively in December 2023.
- Airside equipment electricity consumption for December 2023 has increased by 11.9% compared to the same month in 2022.
- Hotel towers electricity consumption for December 2023 has increased by 20.6% compared to the same month in 2022.
- Kitchen & restaurant electricity consumption for December 2023 has increased by 26.5% compared to the same month in 2022.
- E-bus electricity consumption for December 2023 has increased by 259.2% compared to the same month in 2022.

Table 22 presents the monthly e-bus electricity consumption in MGM COTAI. The e-bus electricity consumption in December 2023 has increased by 259.2% compared to the same month in 2022.

*Table 22. COTAI - Summary of Monthly e-bus Electricity Consumption*

| Monthly Period | E-bus Electricity Consumption (kWh) | | |
|---|---|---|---|
| | 2022 | 2023 | % Change – Actual |
| **January** | 42,287 | 51,261 | +21.2% |
| **February** | 47,905 | 53,640 | +12.0% |
| **March** | 38,793 | 66,865 | +72.4% |
| **April** | 38,537 | 72,669 | +88.6% |
| **May** | 51,328 | 135,032 | +163.1% |
| **June** | 36,420 | 140,991 | +287.1% |
| **July** | - | 148,375 | - |
| **August** | 31,160 | 141,507 | +354.1% |
| **September** | 35,281 | 140,705 | +298.8% |
| **October** | 36,433 | 139,616 | +283.2% |
| **November** | 27,466 | 135,280 | +392.5% |
| **December** | 35,422 | 127,228 | +259.2% |
| **Total/Average** | **421,031** | **1,353,169** | **+221.4%** |

Table 23 presents the comparison of electricity rates in the years 2022 and 2023. The electricity rate in December 2023 decreased by 1.5% compared to the same month in 2022.

*Table 23. COTAI - Summary of Actual Monthly Electricity Rate*

| Monthly Period | Year 2022 – Electricity Rate (MOP/kWh) | Year 2023 – Electricity Rate (MOP/kWh) | % Change – Electricity Rate |
|---|---|---|---|
| **January** | 1.20 | 1.24 | +3.3% |
| **February** | 1.20 | 1.24 | +3.3% |
| **March** | 1.20 | 1.24 | +3.2% |
| **April** | 1.25 | 1.23 | -1.9% |
| **May** | 1.25 | 1.23 | -1.6% |
| **June** | 1.25 | 1.23 | -1.8% |
| **July** | 1.25 | 1.23 | -1.6% |
| **August** | 1.25 | 1.23 | -1.9% |
| **September** | 1.25 | 1.23 | -1.8% |
| **October** | 1.24 | 1.22 | -1.8% |
| **November** | 1.24 | 1.22 | -1.6% |
| **December** | 1.24 | 1.22 | -1.5% |
| **Total/Average** | **1.24** | **1.23** | **-0.5%** |

Table 24 summarizes the monthly normalized electricity consumption. The actual electricity consumption in December 2023 decreased by 5.4% compared to 2019 normalized consumption in the same month.

*Table 24. COTAI - Summary of Normalized Monthly Electricity Consumption*

| Monthly Period | 2019 Normalized Electricity Consumption (kWh) | 2023 Actual Electricity Consumption (kWh) | % Change – Normalized |
|---|---|---|---|
| **January** | 8,209,410 | 8,471,280 | -3.8% |
| **February** | 8,459,577 | 7,869,600 | -12.4% |
| **March** | 9,308,184 | 9,128,520 | -6.8% |
| **April** | 9,880,166 | 9,638,280 | -6.7% |
| **May** | 11,054,334 | 10,993,680 | -4.2% |
| **June** | 11,822,706 | 11,802,600 | -3.1% |
| **July** | 12,284,884 | 12,146,160 | -3.5% |
| **August** | 12,183,423 | 11,961,480 | -4.0% |
| **September** | 11,814,705 | 11,428,920 | -5.7% |
| **October** | 11,404,015 | 10,939,680 | -6.1% |
| **November** | 10,260,479 | 9,667,440 | -8.4% |
| **December** | 9,437,101 | 9,325,680 | -5.4% |
| **Total/Average** | **126,118,982** | **123,373,320** | **-5.7%** |

## 3.2 NATURAL GAS CONSUMPTION

The actual natural gas consumption and cost in December 2023 increased by 39.7% and 52.4% respectively compared to the same month in 2022, respectively. To date, MGM COTAI has increased actual natural gas consumption by 50.5% and associated costs by 62.5%.

*Table 25. COTAI - Summary of Actual Monthly Natural Gas Consumption and Cost*

| Monthly Period | Actual NG Consumption ($m^3$) | | | Actual NG Cost (MOP) | | | % Change – Min OAT | % Change – Food Covers |
|---|---|---|---|---|---|---|---|---|
| | 2022 | 2023 | % Change Actual | 2022 | 2023 | % Change Actual | | |
| **January** | 59,378 | 64,043 | +7.9% | 376,240 | 405,800 | +7.9% | -13.7% | +67.3% |
| **February** | 76,089 | 58,053 | -23.7% | 482,130 | 378,840 | -21.4% | +105.6% | +29.8% |
| **March** | 52,146 | 63,461 | +21.7% | 330,410 | 438,660 | +32.8% | +5.7% | +135.0% |
| **April** | 43,900 | 62,658 | +42.7% | 278,170 | 433,120 | +55.7% | +42.3% | +184.5% |
| **May** | 45,632 | 66,163 | +45.0% | 289,140 | 457,350 | +58.2% | +15.4% | +197.2% |
| **June** | 33,949 | 65,456 | +92.8% | 215,120 | 452,450 | +110.3% | +2.8% | +373.6% |
| **July** | 3,041 | 69,467 | +2184.3% | 19,270 | 480,190 | +2391.9% | +2.7% | +3870.4% |
| **August** | 37,574 | 69,963 | +86.2% | 238,080 | 483,610 | +103.1% | +4.7% | +322.5% |
| **September** | 48,351 | 69,610 | +44.0% | 306,370 | 481,170 | +57.1% | -0.8% | +267.0% |
| **October** | 49,413 | 81,130 | +64.2% | 313,100 | 560,800 | +79.1% | +3.6% | +230.7% |
| **November** | 43,924 | 79,032 | +79.9% | 278,320 | 546,300 | +96.3% | -6.0% | +359.4% |
| **December** | 59,825 | 83,551 | +39.7% | 379,080 | 577,540 | +52.4% | -16.7% | +327.8% |
| **Total/Average** | **553,222** | **832,587** | **+50.5%** | **3,505,430** | **5,695,830** | **+62.5%** | **+7.2%** | **+221.4%** |

***Important note:*** *Minimum Outdoor Air Temperature (OAT is the main variable taken into consideration on natural gas consumption normalization*

NV5 summarized the key items which could result in natural gas consumption increase below:

- Kitchen NG consumption for December 2023 has increased by 68.5% compared to the same month in 2022. Food cover has increased by 327.8% in December 2023.

Table 26 presents the comparison of natural gas rates in the years 2022 and 2023. The natural gas rate in December 2023 increased by **9.1%** compared to the same month in 2022.

*Table 26. COTAI - Summary of Actual Monthly Natural Gas Rate*

| Monthly Period | Year 2022 – Natural Gas Rate (MOP/m³) | Year 2023 – Natural Gas Rate (MOP/m³) | % Change – NG Rate |
|---|---|---|---|
| **January** | 6.34 | 6.34 | - |
| **February** | 6.34 | 6.53 | +3.0% |
| **March** | 6.34 | 6.91 | +9.1% |
| **April** | 6.34 | 6.91 | +9.1% |
| **May** | 6.34 | 6.91 | +9.1% |
| **June** | 6.34 | 6.91 | +9.1% |
| **July** | 6.34 | 6.91 | +9.1% |
| **August** | 6.34 | 6.91 | +9.1% |
| **September** | 6.34 | 6.91 | +9.1% |
| **October** | 6.34 | 6.91 | +9.1% |
| **November** | 6.34 | 6.91 | +9.1% |
| **December** | 6.34 | 6.91 | +9.1% |
| **Total/Average** | **6.34** | **6.83** | **+7.8%** |

Table 27 summarizes the monthly normalized natural gas consumption. The actual natural gas consumption in December 2023 decreased by **35.7%** compared to 2019 normalized consumption in the same month.

*Table 27. COTAI - Summary of Normalized Monthly Natural Gas Consumption*

| Monthly Period | 2019 Normalized Natural Gas Consumption (m³) | 2023 Actual Natural Gas Consumption (m³) | % Change – Normalized |
|---|---|---|---|
| **January** | 128,174 | 64,043 | -50.0% |
| **February** | 120,801 | 58,053 | -51.9% |
| **March** | 117,977 | 63,461 | -46.2% |
| **April** | 113,428 | 62,658 | -44.8% |
| **May** | 112,330 | 66,163 | -41.1% |
| **June** | 104,173 | 65,456 | -37.2% |
| **July** | 102,604 | 69,467 | -32.3% |
| **August** | 101,976 | 69,963 | -31.4% |
| **September** | 105,271 | 69,610 | -33.9% |
| **October** | 112,330 | 81,130 | -27.8% |
| **November** | 117,036 | 79,032 | -32.5% |
| **December** | 129,899 | 83,551 | -35.7% |
| **Total/Average** | **1,365,999** | **832,587** | **-39.0%** |

## 3.3 ISO50001 ENERGY CONSUMPTION RECORD

COTAI ISO50001 energy target is to achieve 13% normalized energy consumption reduction by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to December 2023 is **5.7%** as shown in Table 28 and Table 29 below summarizes the site monthly energy consumption (Electricity and Natural Gas combined) compared to the 2019 baseline.

*Table 28. COTAI ISO 50001 Normalized Energy Performance Targets for 2025*

| MGM Properties | Baseline | Energy Savings | Normalized Savings |
|---|---|---|---|
| COTAI | 2019 | 2025 Target | **-13%** |
| | | Savings as of Dec 2023 | **-5.7%** |

*Table 29. COTAI ISO 50001 Normalized Monthly Energy Consumption*

| Monthly Period | 2019 Normalized Energy Consumption (GJ) | 2023 Actual Energy Consumption (GJ) | % Change – Normalized |
|---|---|---|---|
| **January** | 34,002 | 32,719 | -3.8% |
| **February** | 34,646 | 30,345 | -12.4% |
| **March** | 37,603 | 35,065 | -6.8% |
| **April** | 39,505 | 36,872 | -6.7% |
| **May** | 43,693 | 41,873 | -4.2% |
| **June** | 46,177 | 44,761 | -3.1% |
| **July** | 47,786 | 46,137 | -3.5% |
| **August** | 47,399 | 45,489 | -4.0% |
| **September** | 46,186 | 43,560 | -5.7% |
| **October** | 44,952 | 42,198 | -6.1% |
| **November** | 40,999 | 37,545 | -8.4% |
| **December** | 38,481 | 36,472 | -5.2% |
| **Total/Average** | **501,428** | **473,035** | **-5.7%** |

***Important note**: Factors used for unit conversion: Electricity (0.0036) and Natural Gas (0.0347)*

## 3.4 WATER CONSUMPTION

According to Table 30, the actual water consumption and associated costs in December 2023 has increased by 56.8% and 56.6% respectively compared to the same month in 2022 respectively. To date, MGM COTAI has changed both the actual water consumption and associated cost by 55.8% and 55.6% respectively.

*Table 30. COTAI - Summary of Actual Monthly Water Consumption and Cost*

| Monthly Period | Actual Water Consumption ($m^3$) | | | Actual Water Cost (MOP) | | | Normalization Factors | |
|---|---|---|---|---|---|---|---|---|
| | 2022 | 2023 | % Change Actual | 2022 | 2023 | % Change Actual Cost | % Change – CDD | % Change – Food Covers |
| **January** | 44,658 | 54,018 | +21.0% | 347,240 | 419,790 | +20.9% | -37.7% | +67.3% |
| **February** | 45,298 | 61,317 | +35.4% | 352,330 | 476,420 | +35.2% | +658.0% | +29.8% |
| **March** | 44,914 | 61,936 | +37.9% | 349,260 | 481,140 | +37.8% | -9.0% | +135.0% |
| **April** | 51,640 | 70,617 | +36.7% | 401,500 | 548,440 | +36.6% | -3.2% | +184.5% |
| **May** | 55,628 | 80,211 | +44.2% | 432,480 | 622,920 | +44.0% | +17.6% | +197.2% |
| **June** | 45,805 | 84,354 | +84.2% | 356,090 | 654,900 | +83.9% | +6.6% | +373.6% |
| **July** | 38,031 | 94,905 | +149.5% | 295,890 | 736,800 | +149.0% | -3.1% | +3870.4% |
| **August** | 49,076 | 85,434 | +74.1% | 381,480 | 663,190 | +73.8% | +3.7% | +322.5% |
| **September** | 55,797 | 83,149 | +49.0% | 433,480 | 645,490 | +48.9% | -9.6% | +267.0% |
| **October** | 51,216 | 79,151 | +54.5% | 398,090 | 614,630 | +54.4% | +3.7% | +230.7% |
| **November** | 52,513 | 77,515 | +47.6% | 408,250 | 602,010 | +47.5% | +3.2% | +359.4% |
| **December** | 40,909 | 64,143 | +56.8% | 318,230 | 498,300 | +56.6% | +2462% | +327.8% |
| **Total/Average** | **575,485** | **896,750** | **+55.8%** | **4,474,320** | **6,964,030** | **+55.6%** | **+5.7%** | **+221.4%** |

***Important note:*** *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

NV5 summarized the key items which could result in water consumption increase below:

- Hotel Towers water consumption for December 2023 has increased by 131.2% compared with the same month in 2022.
- Kitchen water consumption for December 2023 has increased by 13.4% while food cover has increased by 327.8% compared with the same month in 2022.

Table 31 presents the comparison of water rates in the years 2022 and 2023. The water rate in December 2023 has decreased by 0.1% compared to the same month in 2022. The average utility rate of water the average utility rate of water has decreased by 0.1% compared to 2022.

*Table 31. COTAI - Summary of Actual Monthly Water Rates*

| Monthly Period | Year 2022 – Water Rate (MOP/m³) | Year 2023 – Water Rate (MOP/m³) | % Change – Water Rate |
|---|---|---|---|
| **January** | 7.776 | 7.771 | -0.1% |
| **February** | 7.778 | 7.770 | -0.1% |
| **March** | 7.776 | 7.768 | -0.1% |
| **April** | 7.775 | 7.766 | -0.1% |
| **May** | 7.774 | 7.764 | -0.1% |
| **June** | 7.774 | 7.764 | -0.1% |
| **July** | 7.780 | 7.764 | -0.2% |
| **August** | 7.773 | 7.763 | -0.1% |
| **September** | 7.769 | 7.763 | -0.1% |
| **October** | 7.773 | 7.765 | -0.1% |
| **November** | 7.774 | 7.766 | -0.1% |
| **December** | 7.779 | 7.769 | -0.1% |
| **Total/Average** | **7.775** | **7.766** | **-0.1%** |

Table 32 summarized the monthly normalized water consumption. The actual water consumption in December 2023 has decreased by 15.5% compared to the same month in 2022. To date, the average utility rate of water has decreased by 0.2% compared to 2022.

*Table 32. COTAI - Summary of Normalized Monthly Water Consumption*

| Monthly Period | 2019 Normalized Water Consumption (m³) | 2023 Actual Water Consumption (m³) | % Change – Normalized |
|---|---|---|---|
| **January** | 57,914 | 54,018 | -6.7% |
| **February** | 56,894 | 61,317 | +7.8% |
| **March** | 62,913 | 61,936 | -1.6% |
| **April** | 69,294 | 70,617 | +1.9% |
| **May** | 75,963 | 80,211 | +5.6% |
| **June** | 79,803 | 84,354 | +5.7% |
| **July** | 85,984 | 94,905 | +10.4% |
| **August** | 85,464 | 85,434 | -0.04% |
| **September** | 81,731 | 83,149 | +1.7% |
| **October** | 90,940 | 79,151 | -13.0% |
| **November** | 75,501 | 77,515 | +2.7% |
| **December** | 75,931 | 64,143 | -15.5% |
| **Total/Average** | **898,331** | **896,750** | **-0.2%** |

## 3.5 COTAI YEAR 2023 PROJECTS

NV5 had performed analysis and coordinate with MGM to implement the following projects in 2023 and reduce the site energy and water consumption which supports NV5 annual normalized energy savings target.

*Table 33. MGM COTAI Year 2023 Projects*

| Item | Initiatives | Status |
|---|---|---|
| **Electricity** | | |
| 1 | Chiller Plant Controls Optimization (CHWP DP Setpoint Reset, CHWST Setpoint Reset) | On-going |
| 2 | PICCV Implementation 2023 (12 Units) | Postpone to 2024 |
| 3 | KEF Steps Speed Control Optimization<br>(Vista, Ball Room, MDM, Hao Guo, Five Foot Road, Chún, Coast) | Completed |
| 4 | AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2) | On-going |
| 5 | Centralized RO Pump Set Replacement (6 units) | Completed |
| 6 | Booster Pump Set Replacement (2 units) | Completed |
| **Water** | | |
| 7 | Extension of B2 Carpark CDP Water Recycling (Phase 5 to 7) | Completed |
| 8 | Extension of B2 Carpark CDP Water Recycling (Phase 11) | On-going |

# APPENDIX A

## Regression Analysis

The 2019 normalized consumption modelling is remained used for tracking the agreed normalized energy savings target for MGM MACAU and MGM COTAI. The following tables show a summary of the selected independent variables for each utility which are used for the normalized consumption modelling in 2023.

## MGM MACAU

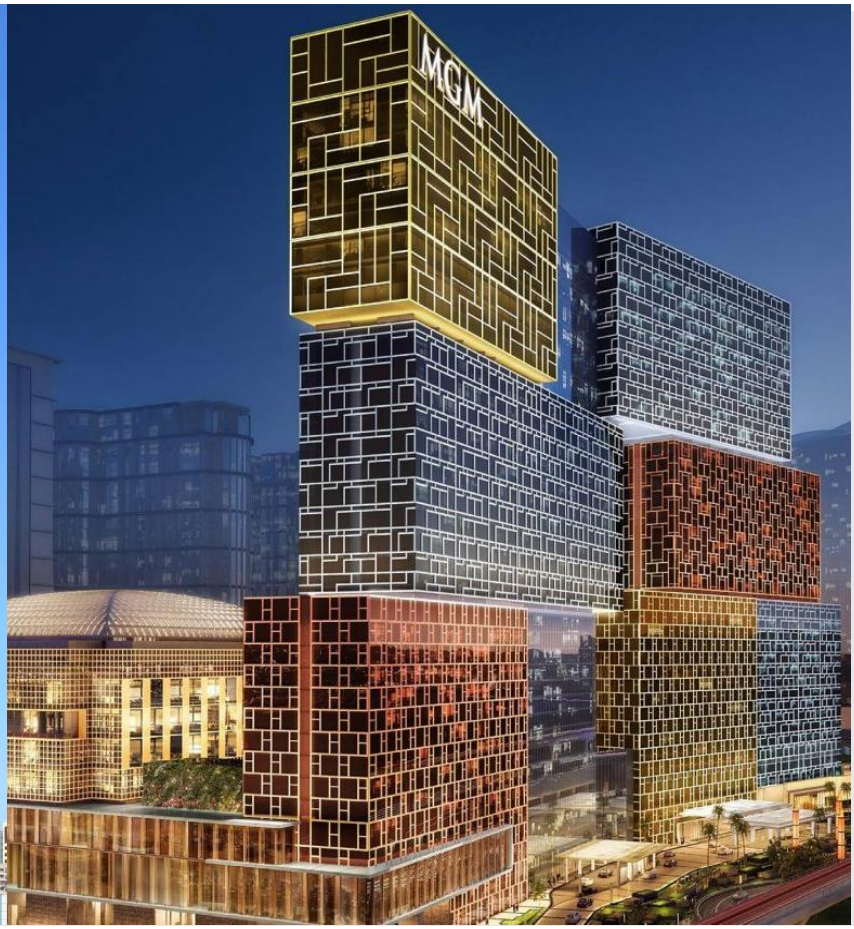
| Utilities | 2023 | |
|---|---|---|
| **Electricity (kWh)** | CDD (Base T=17°C) & Food Cover | |
| **LPG (kg)** | Min OAT | |
| **Water ($m^3$)** | CDD (Base T=17°C) | Food Cover |

## MGM COTAI

| Utilities | 2023 | |
|---|---|---|
| **Electricity (kWh)** | CDD (Base T=17°C), Food Cover,<br><br>Room Night, E-Bus Mileage | |
| **Natural Gas ($m^3$)** | Min OAT | |
| **Water ($m^3$)** | CDD (Base T=17°C) | Food Cover |

# MGM MACAU & COTAI
## Resource Efficiency Management Report
860720-0006650.00
860720-0006651.00




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